GIS Project 2

2107ENV, Trimester 1, 2021

Group 16:

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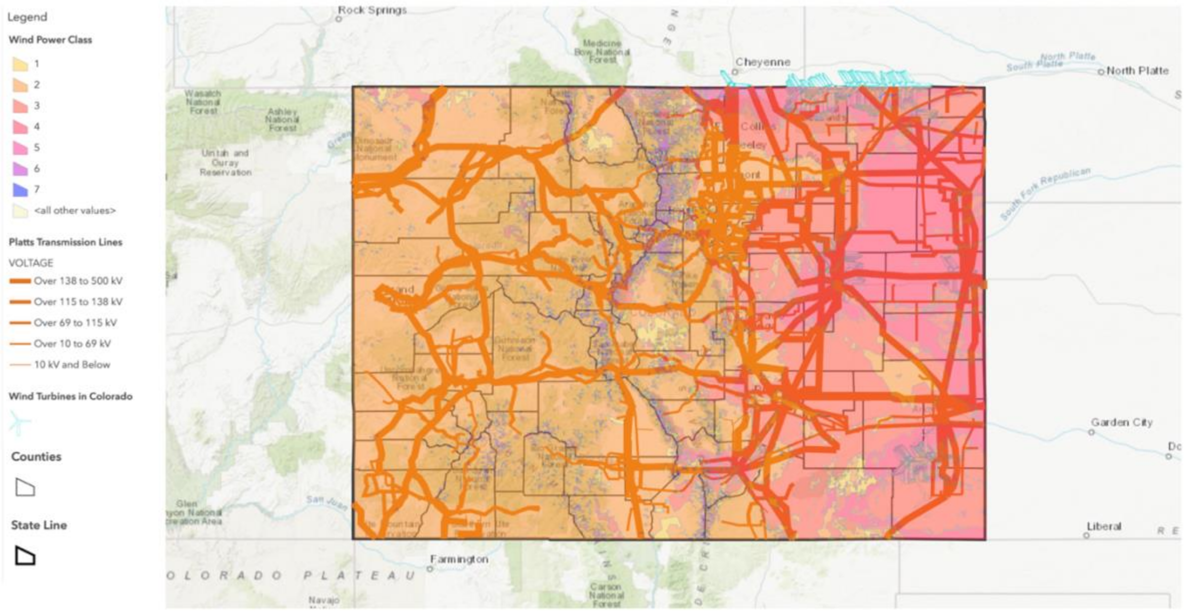
The primary aim of the task was to conduct a site suitability analysis which entailed locating and filtering of potential wind farm sites and thereafter, make a final decision and present this conclusion to the client by means of a functional web application. The results of this investigation and presented below:

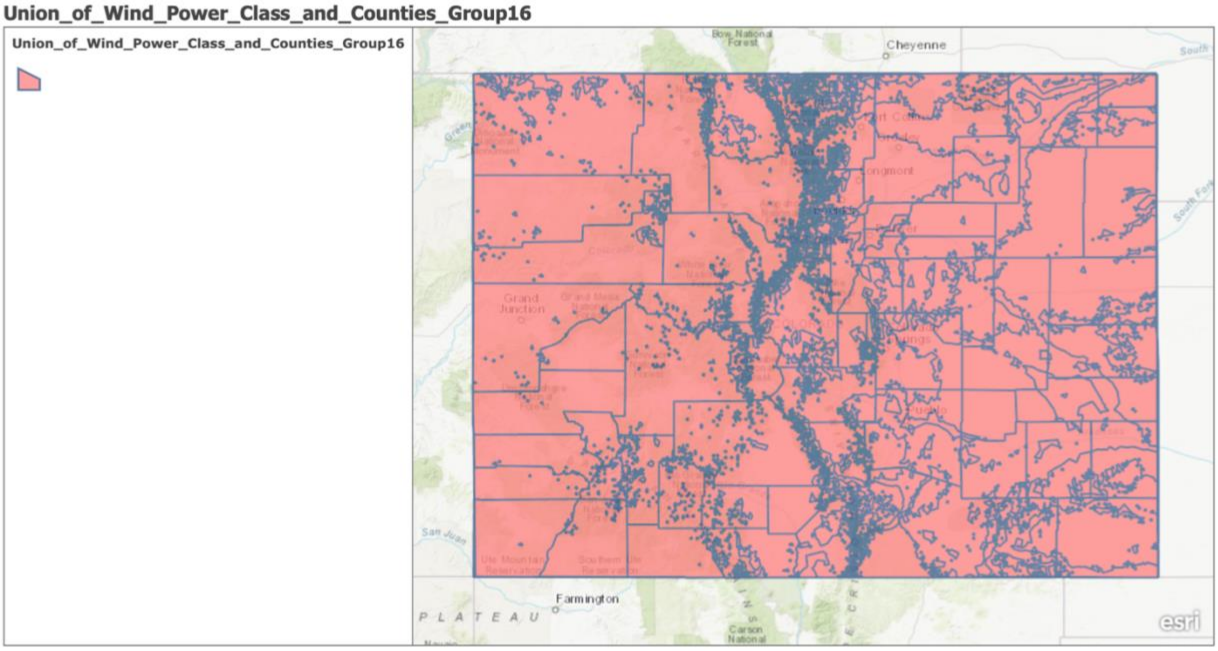
A. <u>Layers for Analysis</u>

1. Initial Map

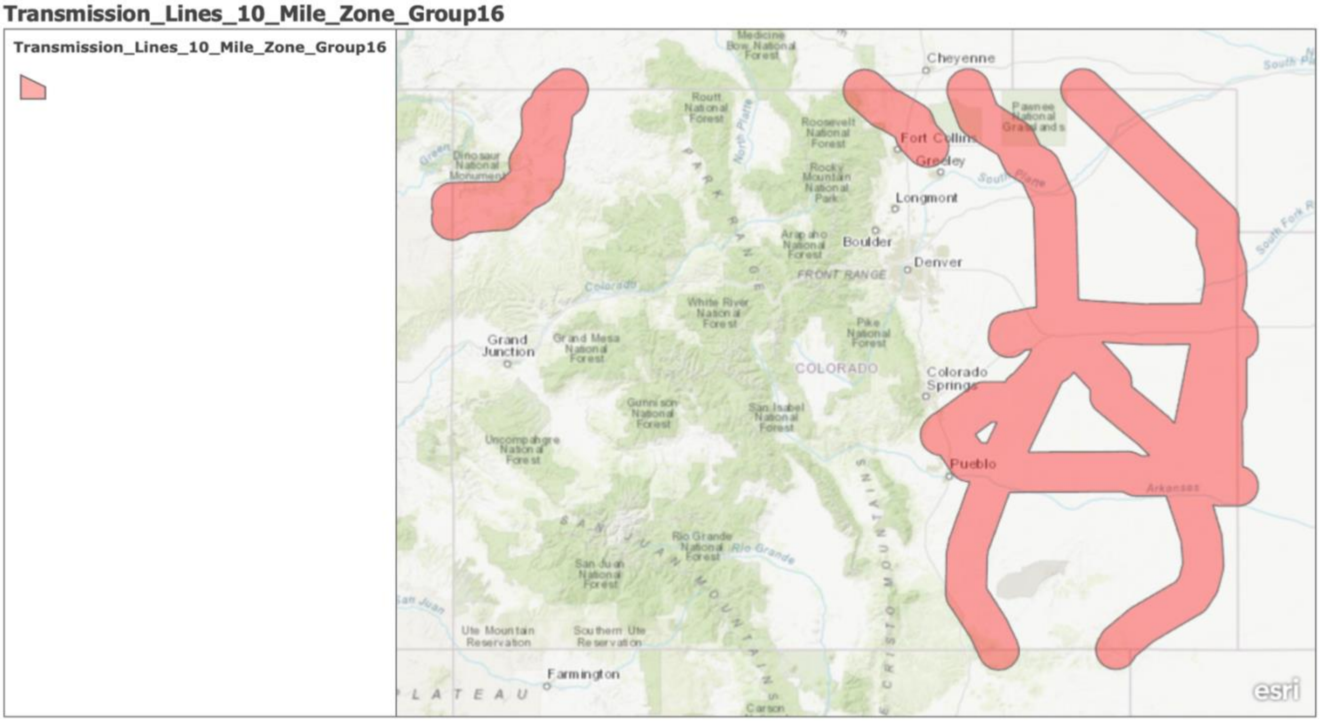


2. Union of Wind Power Class and Countries

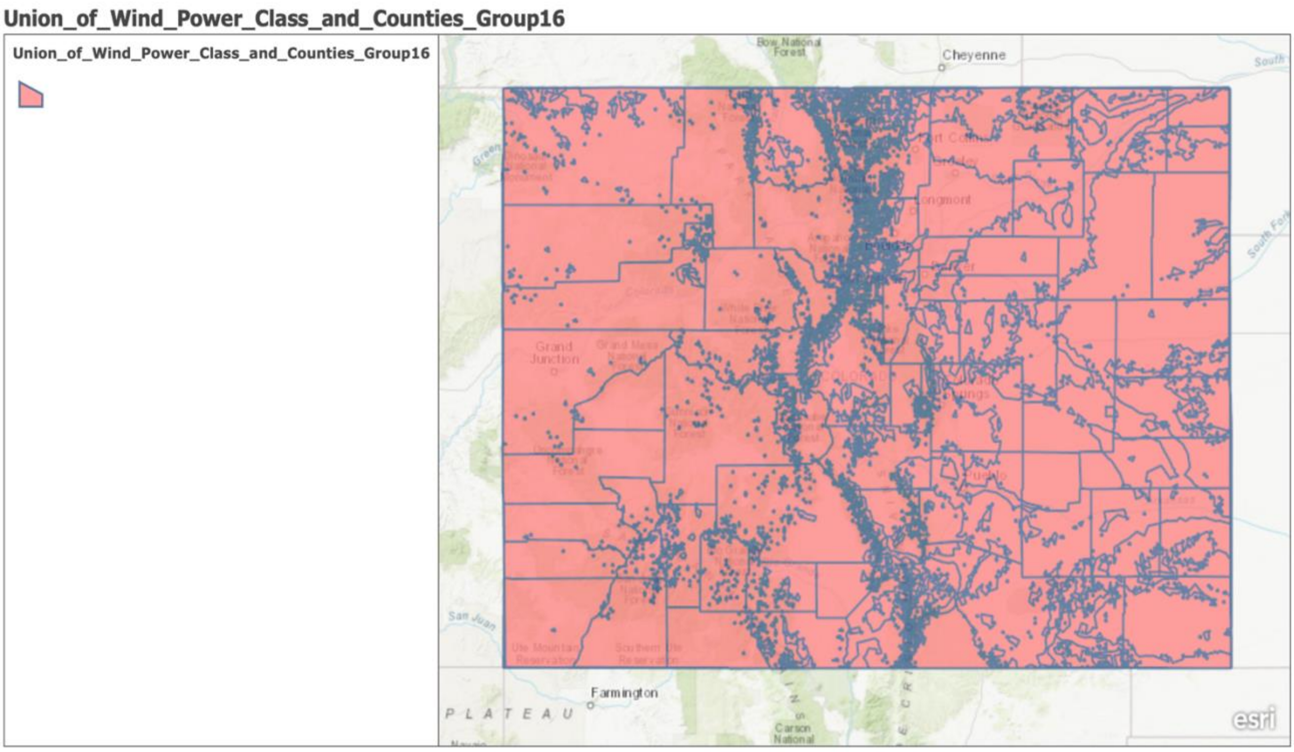
Union_of_Wind_Power_Class_and_Counties_Group16



3. Transmission 10 Mile Zone

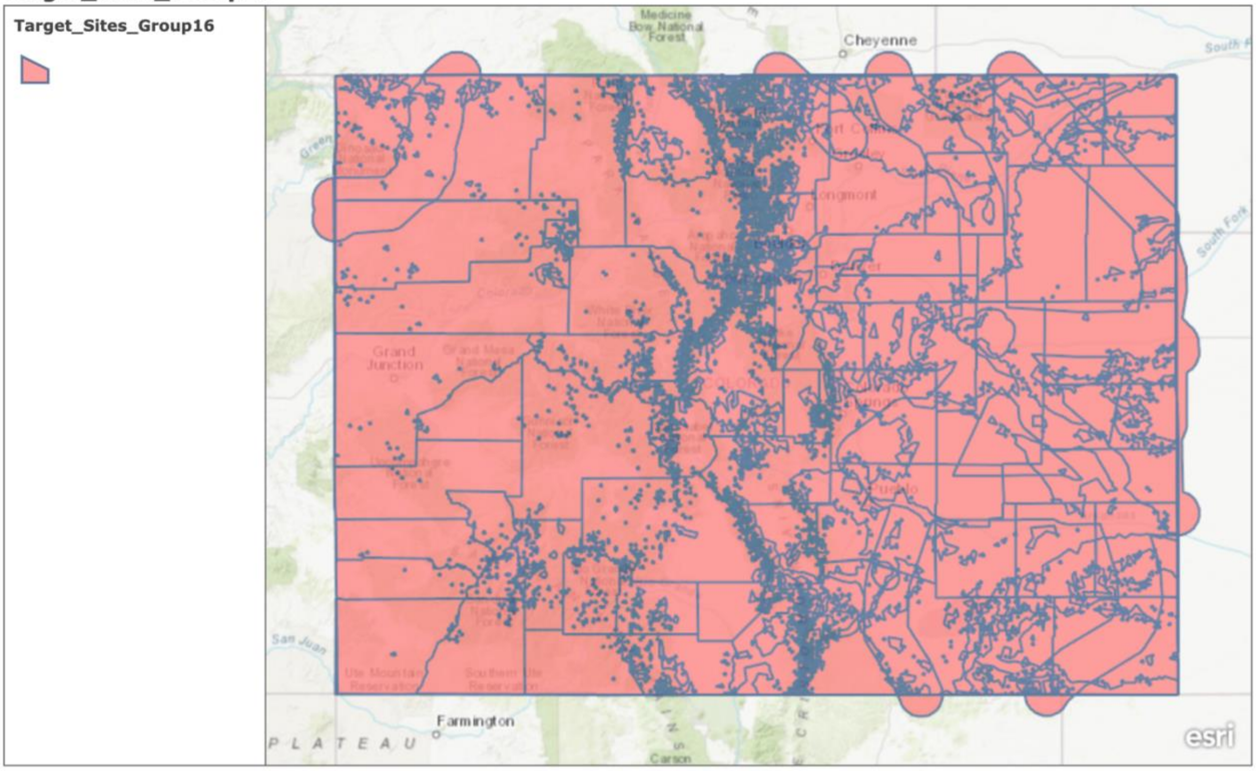


4. Union of Turbine 5 Mile Zone and Transmission Lines 10 Mile Zone



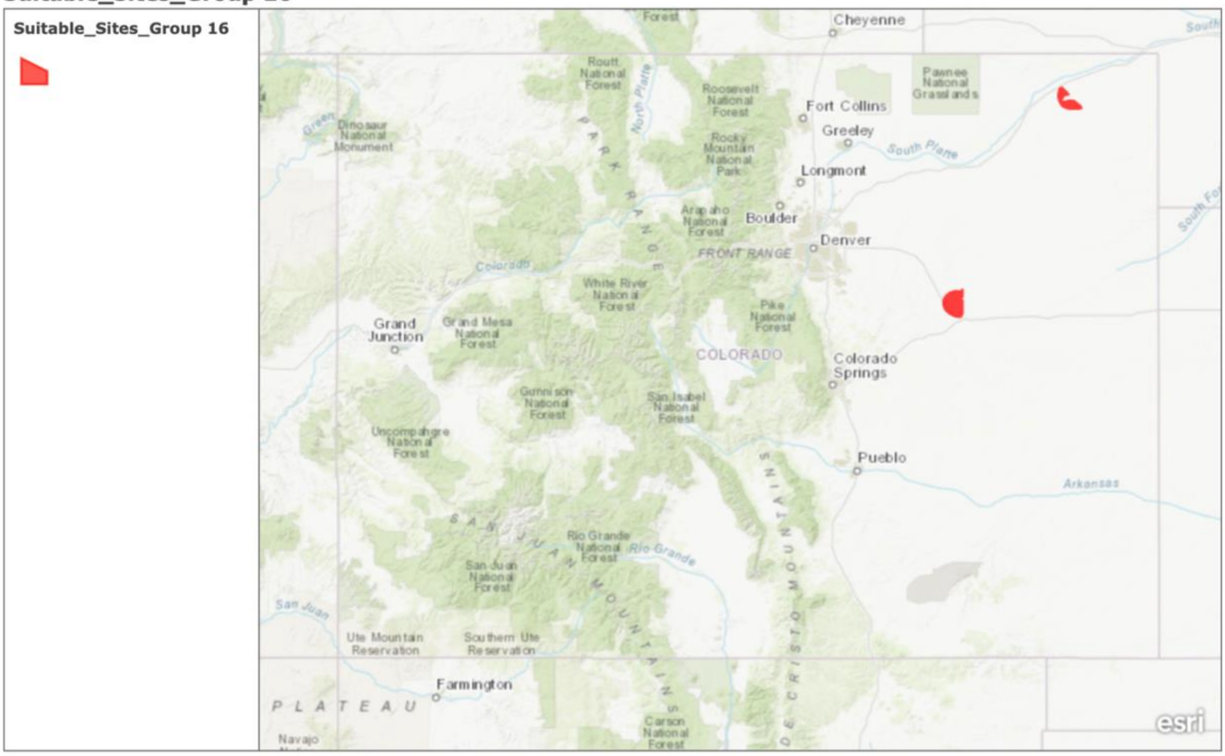
5. Target Sites

Target_Sites_Group16



6. Suitable Sites

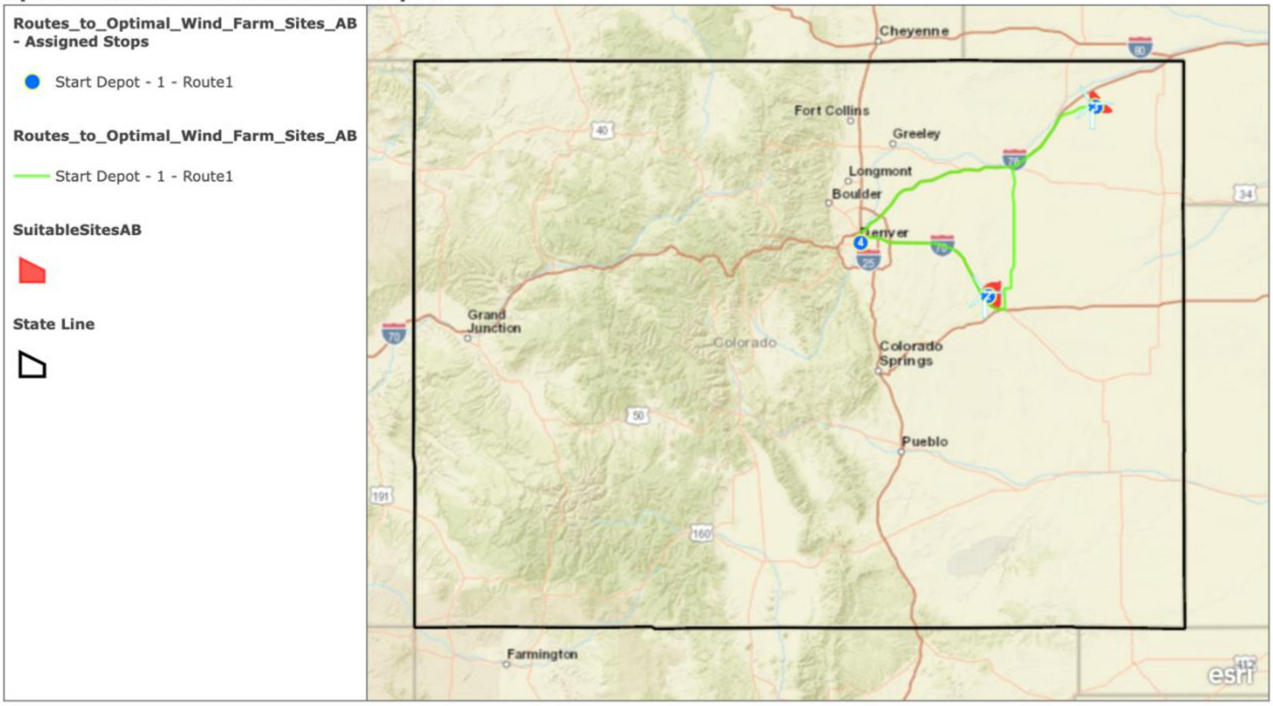
Suitable_Sites_Group 16

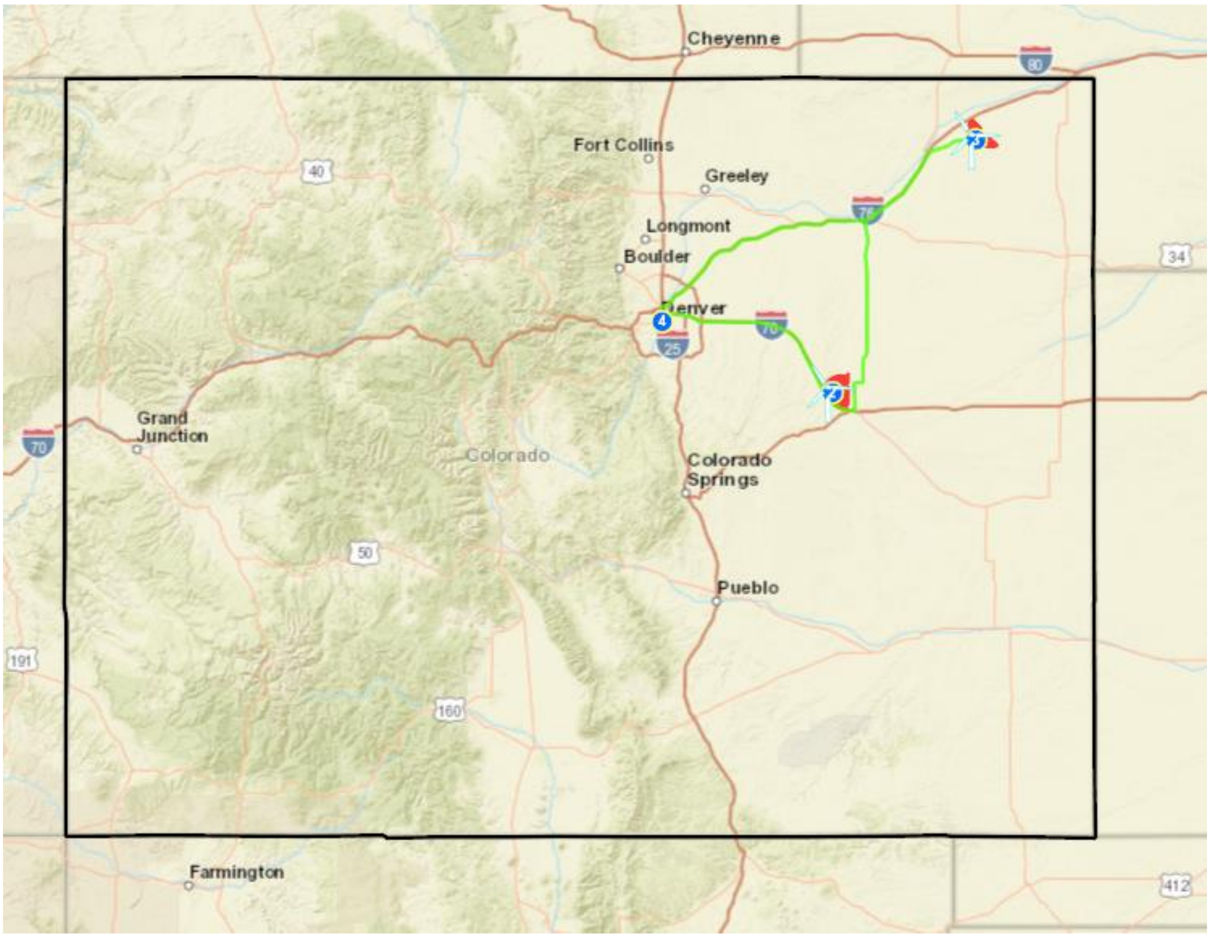


B: Route to Prospective Sites

1. Optimal Wind Farm Sites

Optimum Wind Farm Locations Group 16

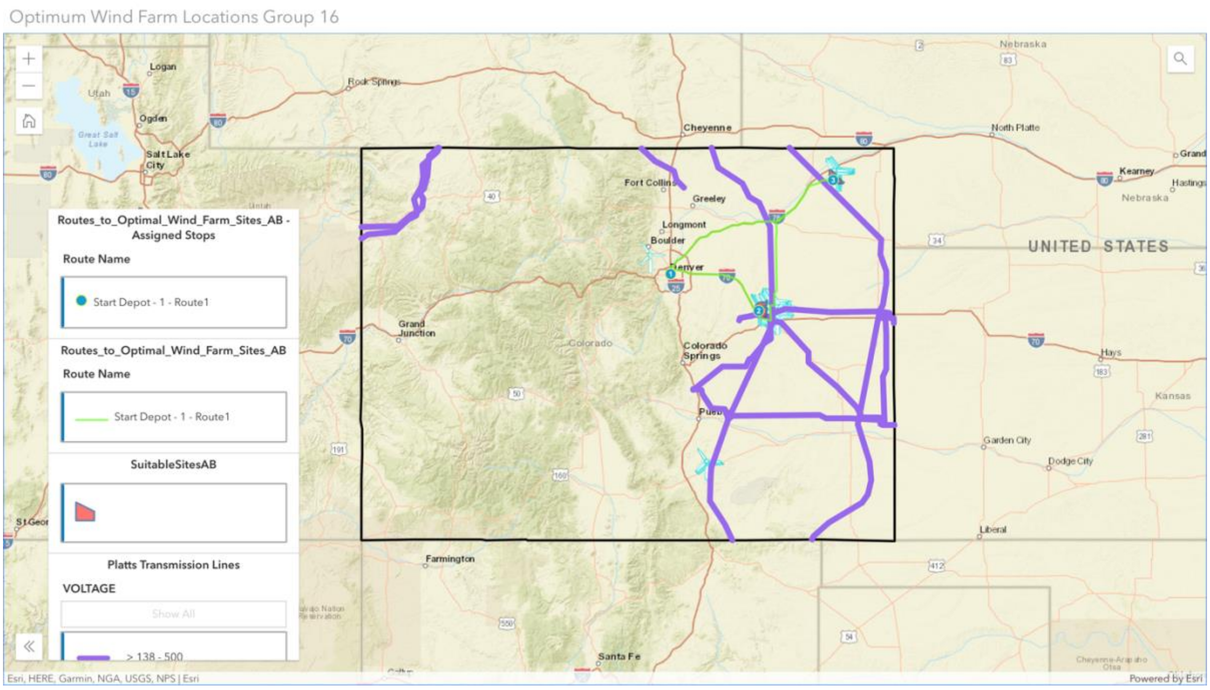




C: Create a Web App

Web App Link:
https://griffith.maps.arcgis.com/apps/instant/interactivelegend/index.html?appid=2cddadccb7ae4ee3b80464a66e7e62f4

Screenshot of the *Interactive Legend* App Interface:



Description

Edit

Wind Turbine Selection Study in Colorado based on the following criteria:

- Located within the state of Colorado.
- In a county where the local or target population is at least 20,000 to ensure an adequate demand.
- Within 10 miles of existing power lines that have a capacity of at least 400 kilovolts (kV), to take advantage of existing energy infrastructure.
- Within 5 miles of existing wind farms containing turbines where the rotor diameters span at least 100 metres.
- Wind power class is at least 4. (Annual wind speeds at 10 metres off the ground are 5.6 metres per second, or 12.5 mph, and at 50 metres off the ground are 7.0 metres per second, or 15.7 mph.)